



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ENGINEERING GRAPHICS AND DESIGN P2

MAY/JUNE 2026

MARKING GUIDELINES

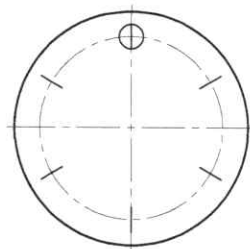
MARKS: 100

These marking guidelines consist of 7 pages.

ANSWERS

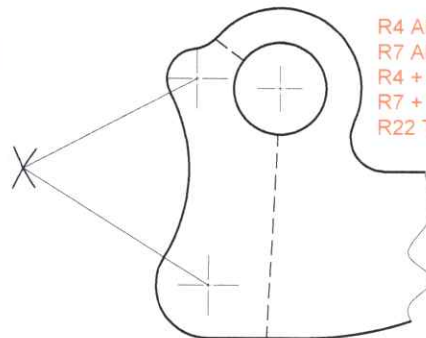
1	<u>APPROVED</u> DRAWINGS	1
2	<u>CHECKED</u> DRAWINGS	1
3	LBJ TRAILER DESIGN	1
4	STEEL	1
5	1 : 5	1
6	11 / ELEVEN	1
7	350 x 2 = <u>700</u>	1
8	SECTIONAL FRONT VIEW	1
9	<u>REVOLVED</u>	1
10	<u>MULTI-PLANE</u> / <u>OFFSET</u>	1
11	<u>INTERRUPTED</u> VIEW	1
12	INDICATES A PART NOT PRESENT	1
13	CONVENTION OF <u>SYMMETRY</u> APPLIED	1
14	SQUARE ON A SHAFT	1
15	F: 11,2 G: 64	2
16	99°/ <u>100°</u> /101°	1
17	33 + 4 + 0,55 ① = 37,55 ①	1
18	See below	3
19		4
20		5
TOTAL		30

ANSWER 18



FREEHAND 0,5
PCD CHAIN LINE 1
ONE HOLE ONLY 0,5
5 x DIVISIONS 1
3

ANSWER 19

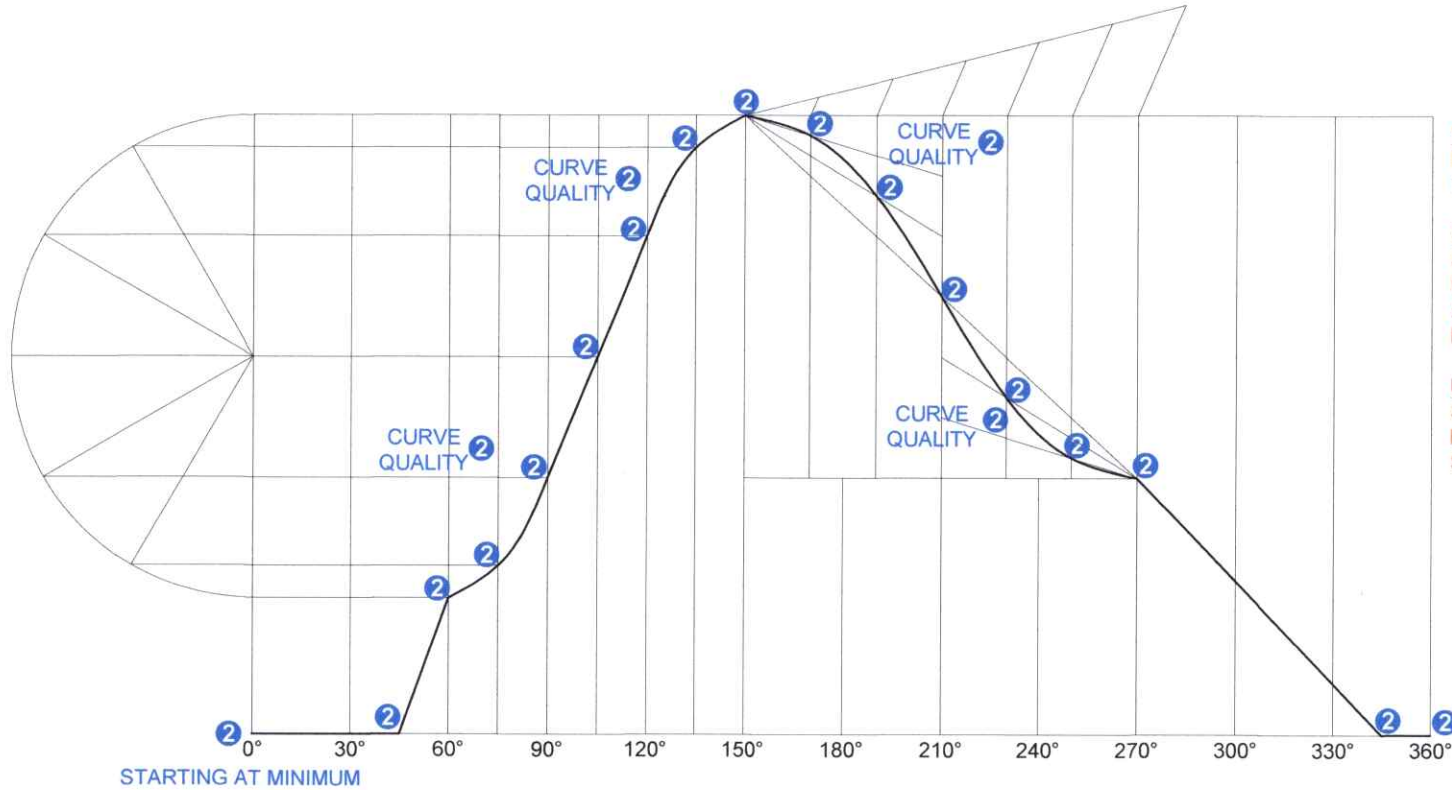


R4 ARC 0,5
R7 ARC 0,5
R4 + R22 = R26 1
R7 + R22 = R29 1
R22 TANGENTIAL ARC 1
4

ANSWER 20

DESCRIPTION		
WELD ALL AROUND	8	✓
PITCH OF WELD	4	✓
FILLET WELD	7	✓
LENGTH OF WELD	3	✓
WELDING PROCESS	5	✓

PAPER 2 QUESTION 1
GRADE 12
MAY/JUNE 2026
MARKING GUIDELINE

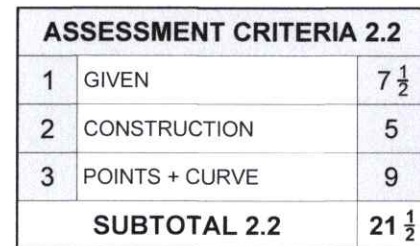


DISPLACEMENT GRAPH 3
SCALE 156 = 360° 3

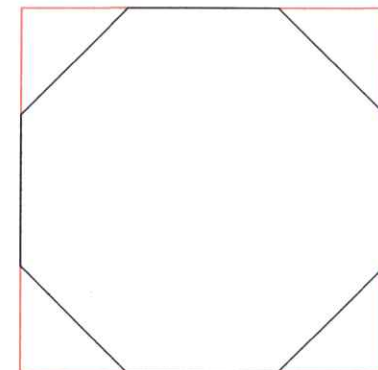
1. GRAPH CONSTRUCTION	
LENGTH 156	1
12 EQUAL DIVISIONS	1
SIMPLE HARMONIC MOTION	
SEMI-CIRCLE IN CORRECT POSITION	0.5
CIRCLE DIVISION	0.5
ADDITIONAL 15° DIVISIONS	0.5
HORIZONTAL PROJECTION	0.5
UNIFORM ACCELERATION AND RETARDATION	
SAME AMOUNT OF V & H DIVISIONS	1
FIRST HALF CONSTRUCTION	0.5
SECOND HALF CONSTRUCTION	0.5
TOTAL	6

ASSESSMENT CRITERIA 2.1		
1	GRAPH CONSTRUCTION	6
2	POINTS + CURVE	10 1/2
3	LABELS	1
SUBTOTAL 2.1		17 1/2

PAPER 2 QUESTION 2.1
GRADE 12
MAY/JUNE 2026
MARKING GUIDELINES



PAPER 2 QUESTION 2.2
GRADE 12
MAY/JUNE 2026
MARKING GUIDELINES

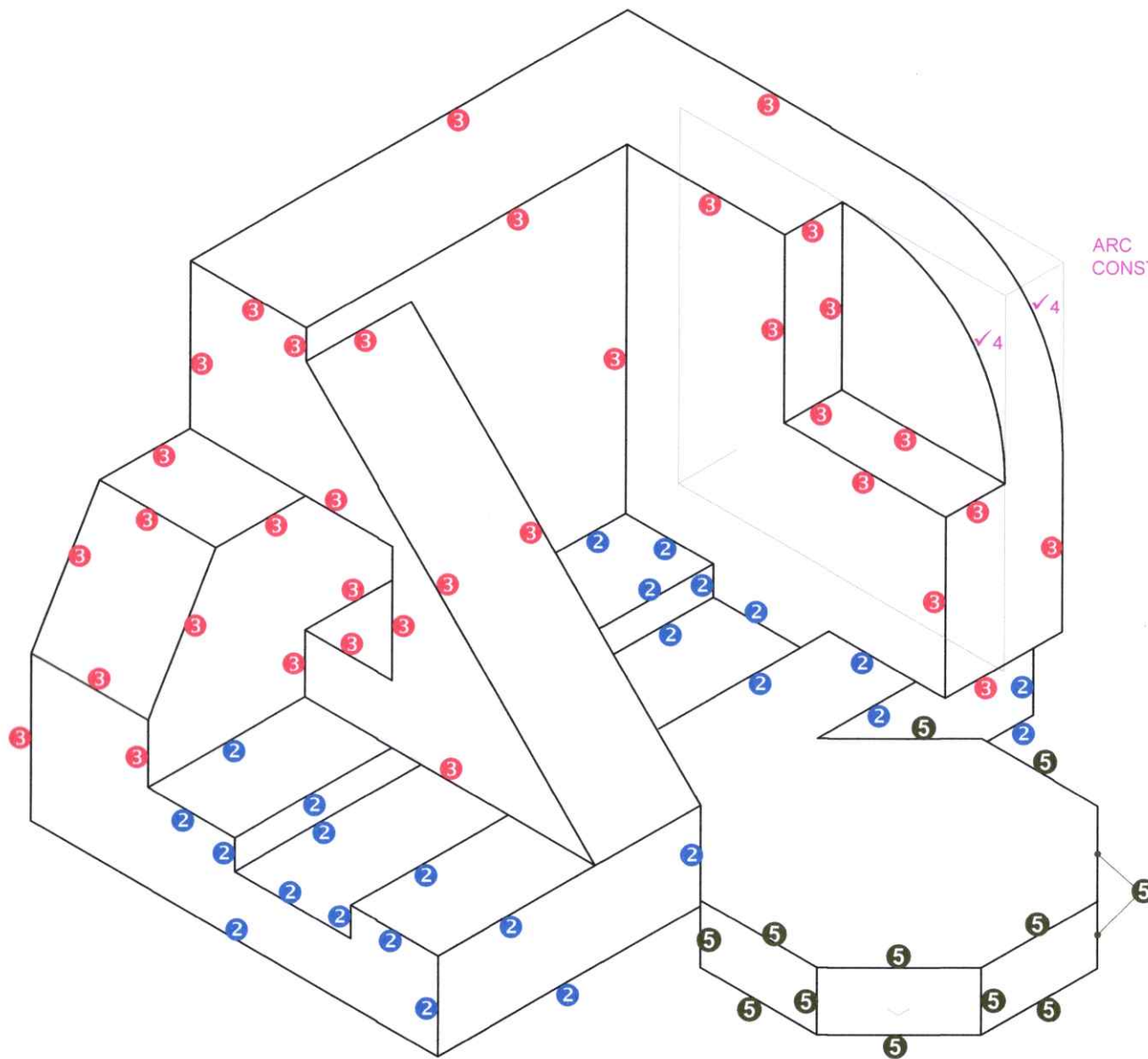


OCTAGON CONSTRUCTION ✓ 1

60° CONSTRUCTION 1



ARC CONSTRUCTION 4



P START 1 / ORIENTATION 1

ASSESSMENT CRITERIA		
1	PLACING + AUX. VIEW	2 ½
2	BASE	12 ½
3	UPPER PORTION	17 ½
4	ARCS + CONSTRUCTION	2 ½
5	OCTAGON	6
TOTAL		41

PAPER 2 QUESTION 3
GRADE 12
MAY/JUNE 2026
MARKING GUIDELINE



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SENIORSERTIFIKAAT-EKSAMEN/ NASIONALE SENIORSERTIFIKAAT-EKSAMEN

INGENIEURSGRAFIKA EN -ONTWERP V2

MEI/JUNIE 2026

NASIENRIGLYNE

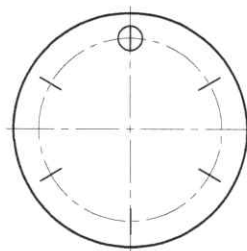
PUNTE: 100

Hierdie nasienriglyne bestaan uit 7 bladsye.

ANTWOORDE

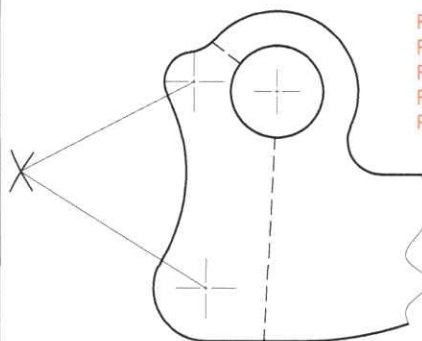
1	TEKENINGE <u>GOEDGEKEUR</u>	1
2	TEKENINGE <u>NAGESIEN</u>	1
3	LBJ SLEEPWA-ONTWERP	1
4	STAAL	1
5	1 : 5	1
6	11 / ELF	1
7	350 x 2 = <u>700</u>	1
8	DEURSNEE-VOORAANSIG	1
9	<u>GEWENTELDE</u>	1
10	<u>MULTI-VLAK</u>	1
11	<u>ONDERBROKE</u> AANSIG	1
12	DUI 'N DENKBEELDIGE ONDERDEEL AAN	1
13	KONVENSIE VAN <u>SIMMETRIE</u> TOEGEPAS	1
14	VIERKANT OP 'N AS	1
15	F: 11,2 G: 64	2
16	99°/ <u>100°</u> /101°	1
17	33 + 4 + 0,55 ① = 37,55 ①	1
18	Sien hieronder	3
19		4
20		5
TOTAAL		30

ANTWOORD 18



VRYHAND 0,5
 SSD KETTINGLYN 1
 SLEGS EEN GAT 0,5
 5 x VERDELINGS $\frac{1}{3}$

ANTWOORD 19

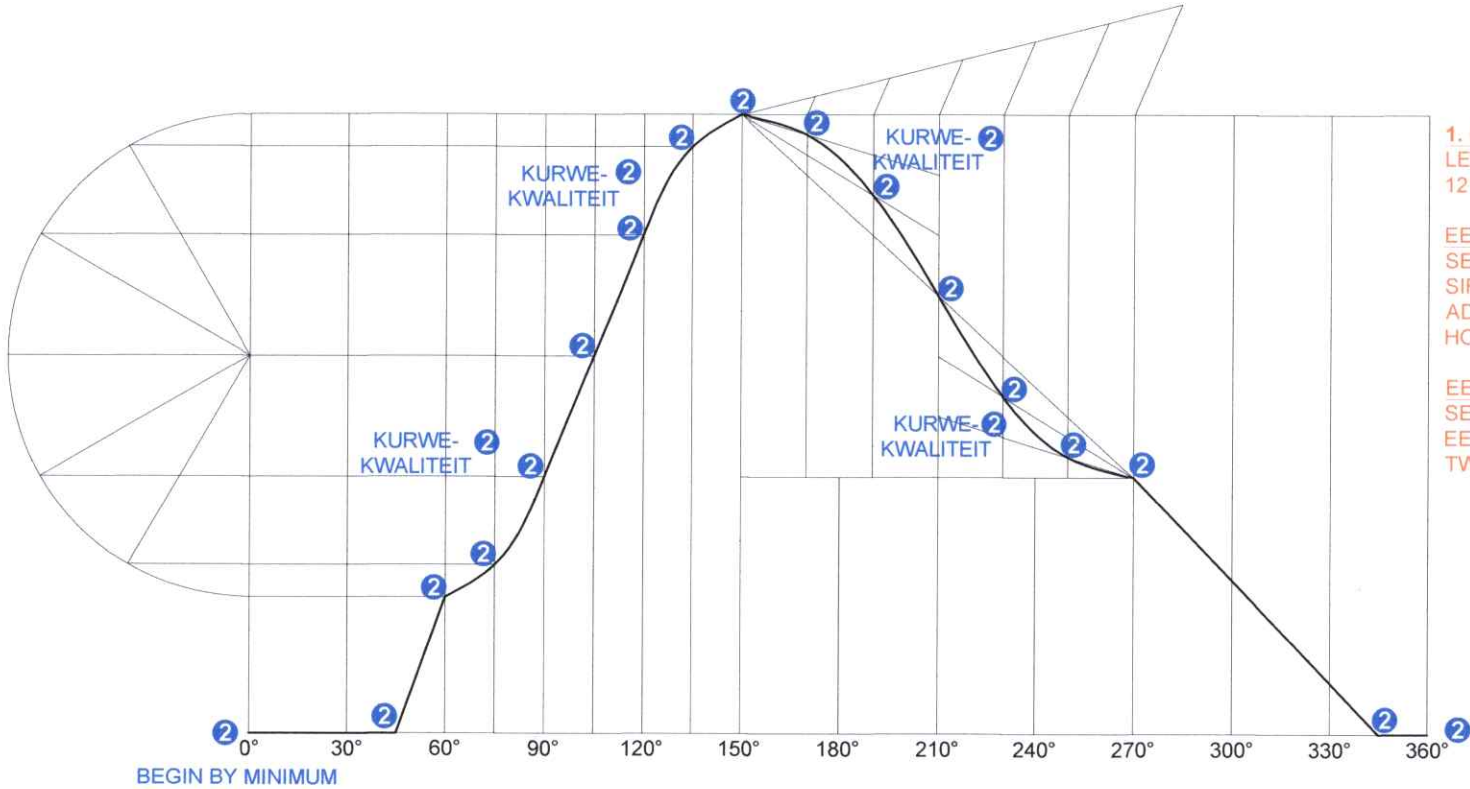


R4 BOOG 0,5
 R7 BOOG 0,5
 R4 + R22 = R26 1
 R7 + R22 = R29 1
 R22 RAAKBOOG $\frac{1}{4}$

BESKRYWING

RONDOMSWEIS	8	✓
SWEISSTEK	4	✓
HOEKSWEISLAS	7	✓
LENGTE VAN SWEISLAS	3	✓
SWEISPROSES	5	✓

VRAESTEL 2 VRAAG 1
 GRAAD 12
 MEI/JUNIE 2026
 NASIENRIGLYNE



VERPLASINGSGRAFIEK ③
SKAAL 156 = 360° ③

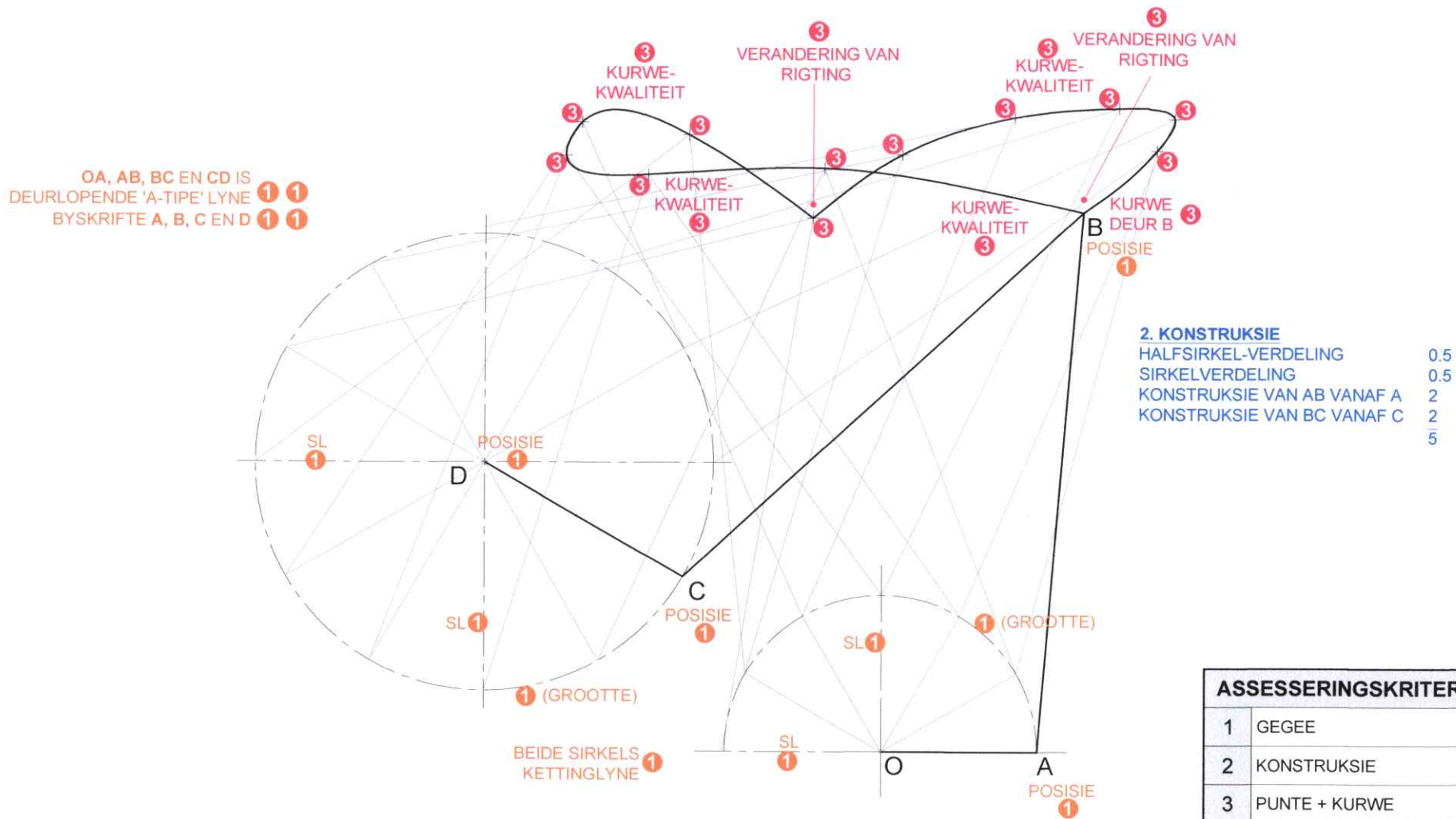
1. GRAFIEKKONSTRUKSIE	
LENGTE 156	1
12 GELYKE INDELINGS	1
EENVOUDIGE HARMONIESE BEWEGING	
SEMI-SIRKEL IN KORREKTE POSISIE	0.5
SIRKELVERDELING	0.5
ADDISIONELE 15°-VERDELINGS	0.5
HORISONTALE PROJEKSIE	0.5
EENVORMIGE VERSNEL. + VERTRAGING	
SELFDE HOEVEELHEID V & H VERDELING	1
EERSTE HELFTE VAN KONSTR.	0.5
TWEEDE HELFTE VAN KONSTR.	0.5
TOTAAL	6

ASSESSERINGSKRITERIA 2.1		
1	GRAFIEKKONSTRUKSIE	6
2	PUNTE + KURWE	10 ¹ / ₂
3	BYSKRIFTE	1
SUBTOTAAL 2.1		17 ¹/₂

VRAESTEL 2 VRAAG 2.1
GRAAD 12
MEI/JUNIE 2026
NASIENRIGLYNE

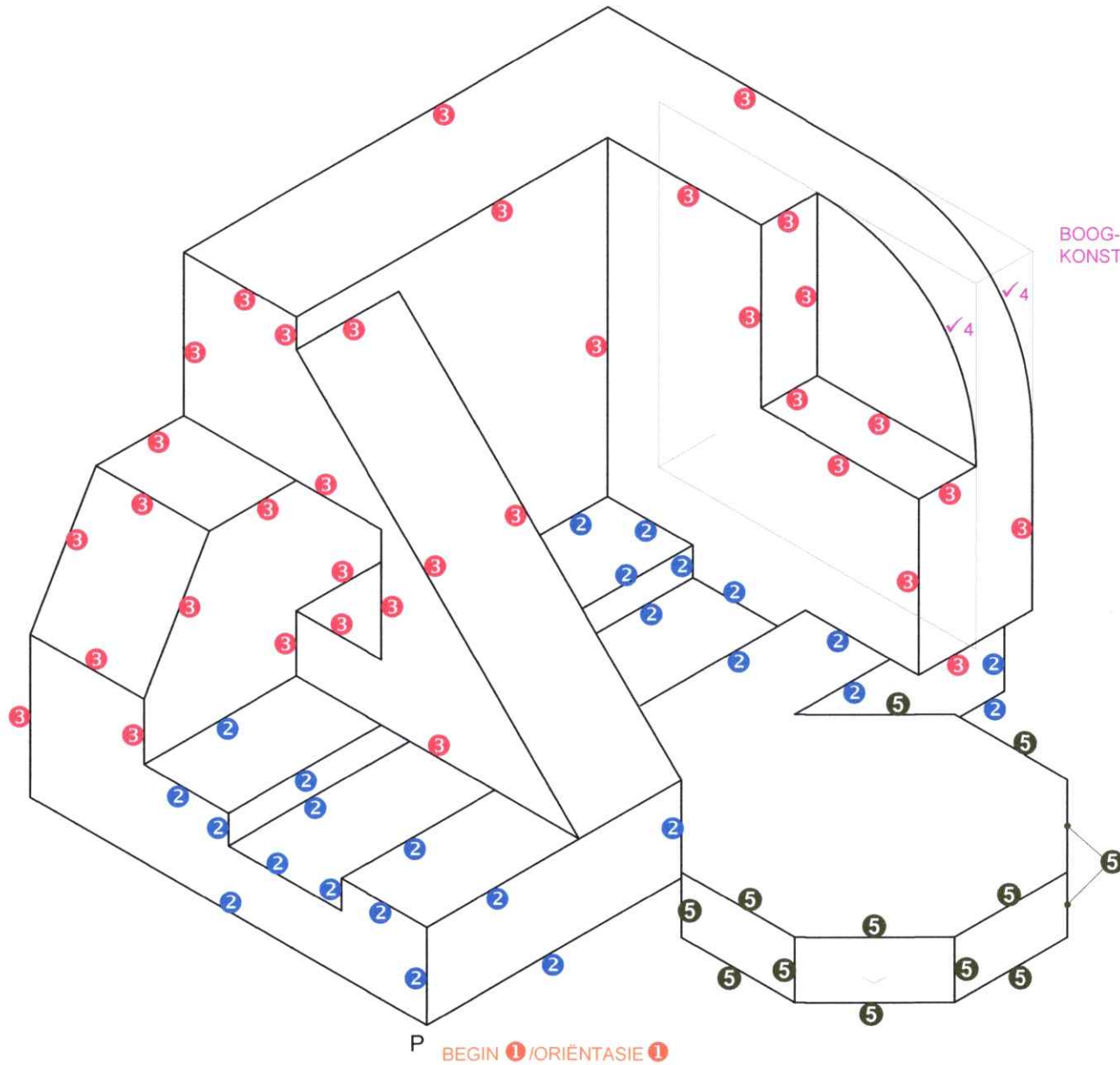
Kopiereg voorbehou

Blaai om asseblief

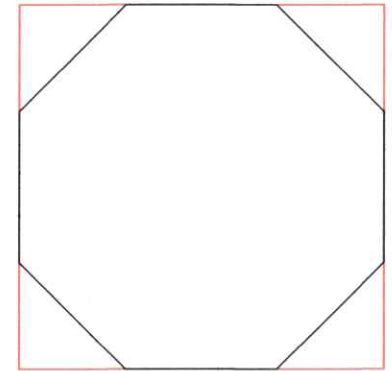


ASSESSERINGSKRITERIA 2.2		
1	GEGEE	7 $\frac{1}{2}$
2	KONSTRUKSIE	5
3	PUNTE + KURWE	9
SUBTOTAAL 2.2		21 $\frac{1}{2}$

VRAESTEL 2 VRAAG 2.2
GRAAD 12
MEI/JUNIE 2026
NASIENRIGLYNE

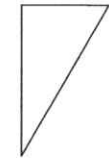


BOOG-KONSTRUKSIE 4



AGTHOEK-KONSTRUKSIE ✓ 1

60°-KONSTRUKSIE 1



ASSESSERINGSKRITERIA		
1	PLASING + HULPAANSIG	2 ½
2	BASIS	12 ½
3	BOONSTE GEDEELTE	17 ½
4	BOË + KONSTRUKSIE	2 ½
5	AGTHOEK	6
TOTAAL		41

VRAESTEL 2 VRAAG 3
GRAAD 12
MEI/JUNIE 2026
NASIENRIGLYNE